Ampere's Law

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Ampere's (Circuital) Law (Similar to Gauss's law)

Ampère's circuital law states that the line integral of **H** about any *closed* path is exactly equal to the direct current enclosed by that path,

$$\oint \mathbf{H} \cdot d\mathbf{L} = I_{encl}$$

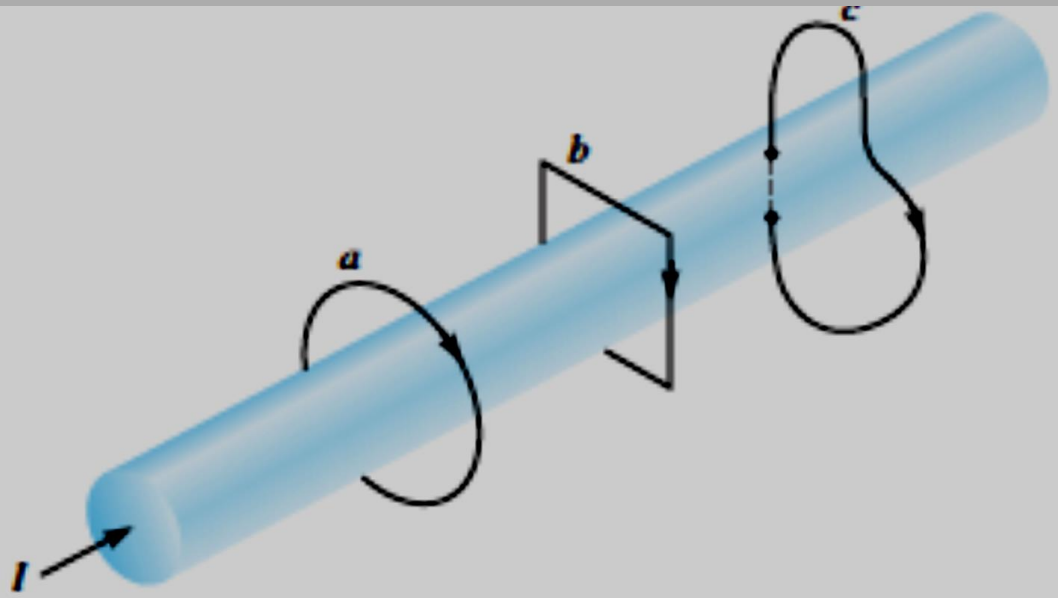


Figure 7.7 A conductor has a total current *I*. The line integral of **H** about the closed paths *a* and *b* is equal to *I*, and the integral around path *c* is less than *I*, since the entire current is not enclosed by the path.

- Find H due to infinitely long current carrying conductor

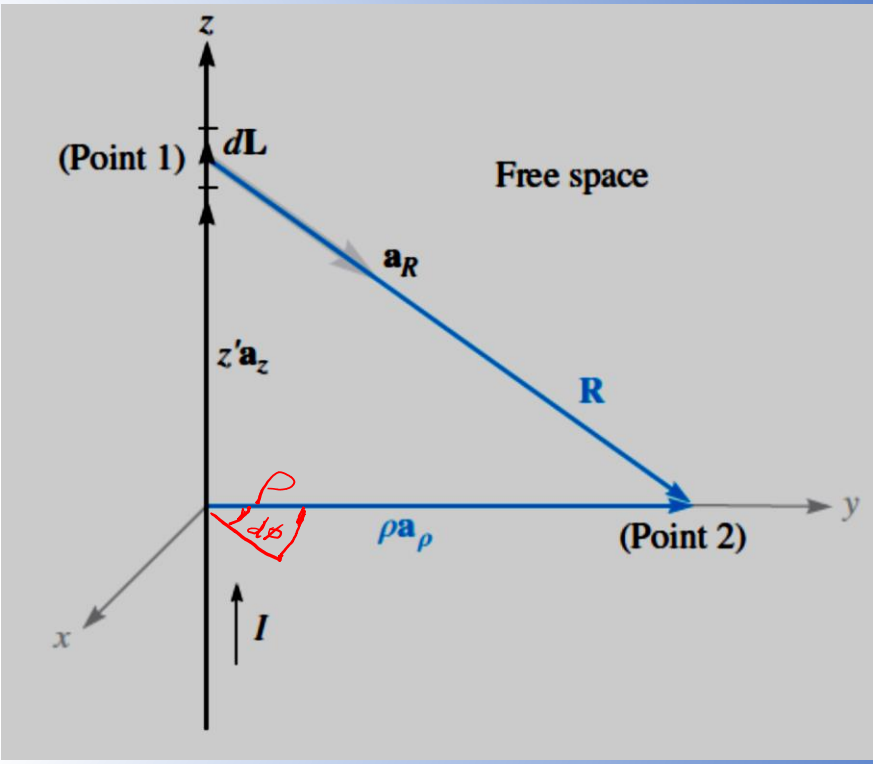
Direction of *d***H** is perpendicular to the plane containing *d***L** and **R** and therefore is in the direction of $\mathbf{a}_\phi$

$$\oint \mathbf{H} \cdot d\mathbf{L} = \int_0^{2\pi} H_\phi \rho d\phi$$

$$= H_\phi \rho \int_0^{2\pi} d\phi$$

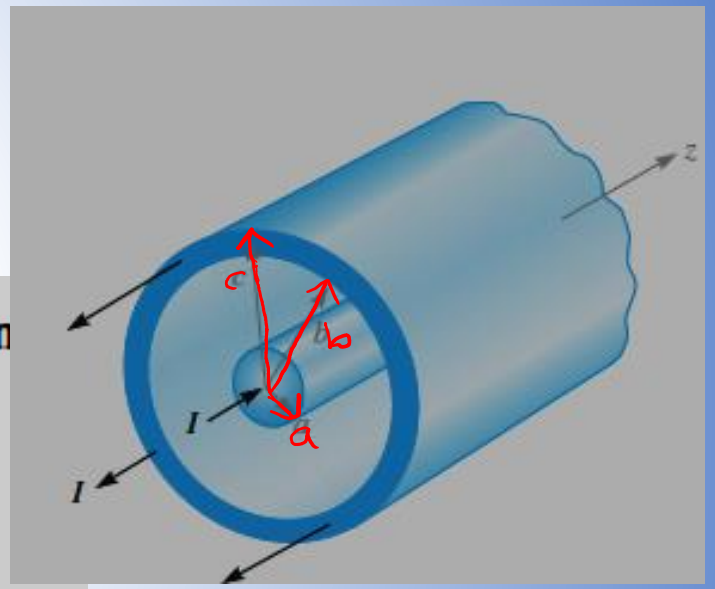
$$= H_\phi 2\pi\rho = I$$

$$H_\phi = \frac{I}{2\pi\rho}$$



Infinitely long coaxial transmission line carrying a uniformly distributed total current I in the center conductor and −I in the outer conductor

$$H_\phi = \frac{I}{2\pi\rho} \quad (a < \rho < b)$$



If we choose ρ smaller than the radius of the inner conductor, the curren
enclosed is

$$I_{encl} = I\frac{\rho^2}{a^2}$$

and

$$2\pi\rho H_\phi = I\frac{\rho^2}{a^2}$$

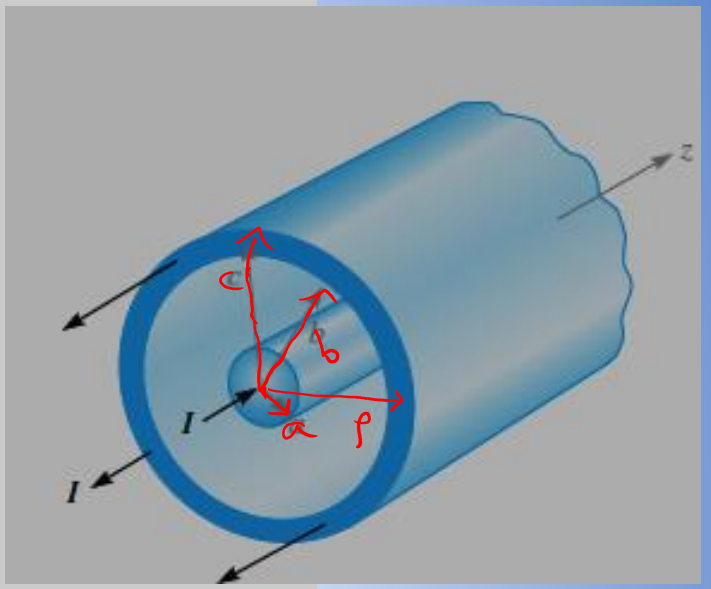
or

$$H_\phi = \frac{I\rho}{2\pi a^2} \quad (\rho < a)$$

If the radius ρ is larger than the outer radius of the outer conductor, no current is enclosed and

$$H_\phi = 0 \quad (\rho > c)$$

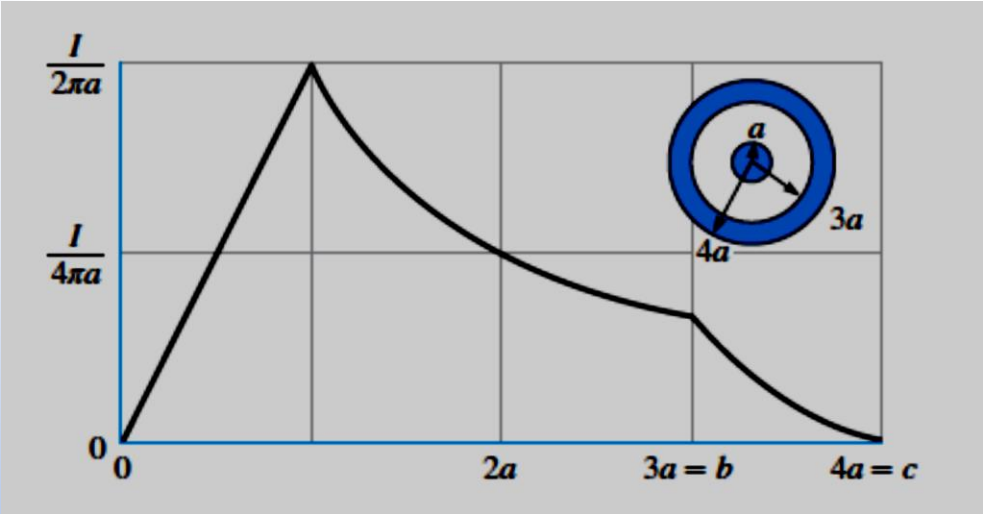
(I - I)



Finally, if the path lies within the outer conductor, we have

$$2\pi\rho H_\phi = I - I\left(\frac{\rho^2 - b^2}{c^2 - b^2}\right)$$

$$H_\phi = \frac{I}{2\pi\rho}\frac{c^2 - \rho^2}{c^2 - b^2} \quad (b < \rho < c)$$



$$H_\phi = \frac{I\rho}{2\pi a^2} \quad (\rho < a)$$

$$H_\phi = \frac{I}{2\pi\rho} \quad (a < \rho < b)$$

Find **H** on the axis of a circular current loop of radius a. Specialize the result to the center of the loop.

For the point shown in Fig. 10-15,

$$\mathbf{R} = -a\mathbf{a}_r + h\mathbf{a}_z$$

$$d\mathbf{H} = \frac{(Ia\,d\phi\,\mathbf{a}_\phi)\times(-a\mathbf{a}_r + h\mathbf{a}_z)}{4\pi(a^2+h^2)^{3/2}}$$

$$= \frac{(Ia\,d\phi)(a\mathbf{a}_z + h\mathbf{a}_r)}{4\pi(a^2+h^2)^{3/2}}$$

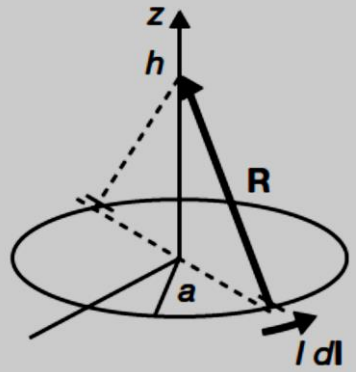


Fig. 10-15

$$\mathbf{H} = \int_0^{2\pi} \frac{Ia^2\,d\phi}{4\pi(a^2+h^2)^{3/2}}\mathbf{a}_z$$

$$= \frac{Ia^2}{2(a^2+h^2)^{3/2}}\mathbf{a}_z$$

At $h = 0$, $\mathbf{H} = (I/2a)\mathbf{a}_z$.

H due to uniform surface current density K_y $\mathbf{a}_y$

- If the sheet is subdivided into a number of filaments, it is evident that no filament can produce an *Hy* component.
- Biot-Savart law shows that the contributions to *Hz* produced by a symmetrically located pair of filaments cancel. Thus, *Hz* is zero also;
- only an *Hx* component is present.

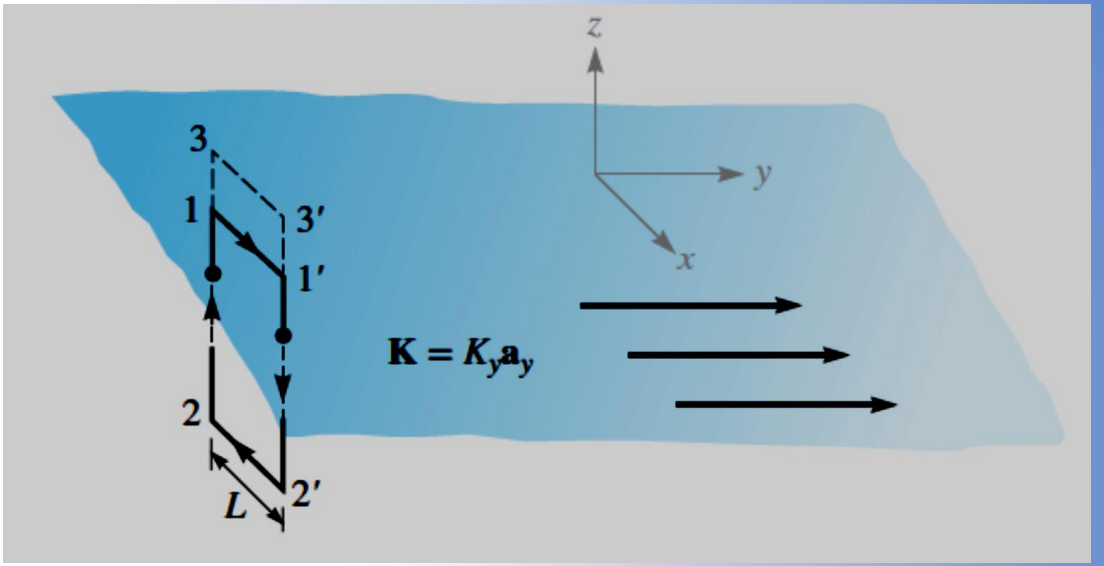
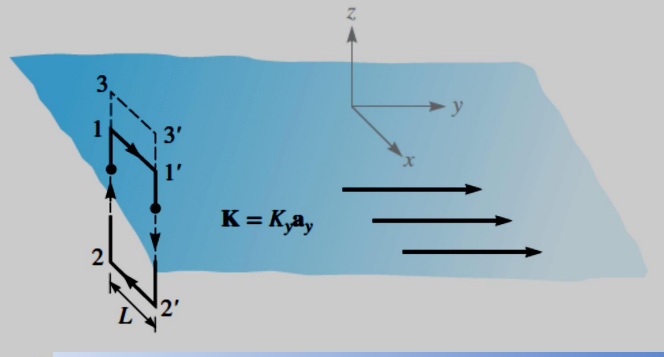


Figure 7.10 A uniform sheet of surface current $\mathbf{K} = K_y\mathbf{a}_y$ in the $z = 0$ plane. **H** may be found by applying Ampère's circuital law about the paths 1-1′-2′-2-1 and 3-3′-2′-2-3.

Using Ampere's law on the path 1-1'-2'-2-1 $\oint \mathbf{H} \cdot d\mathbf{L} = I$



$$H_{x1}L + H_{x2}(-L) = K_y L \quad \text{or} \quad H_{x1} - H_{x2} = K_y$$

If the path 3-3′-2′-2-3 is now chosen, the same current is enclosed, and

$$H_{x3} - H_{x2} = K_y$$

and therefore

$$H_{x3} = H_{x1}$$

It follows that H_x is the same for all positive z. Similarly, H_x is the same for all negative z. Because of the symmetry, then, the magnetic field intensity on one side of the current sheet is the negative of that on the other. Above the sheet,

$$H_x = \tfrac{1}{2}K_y \quad (z > 0)$$

while below it

$$H_x = -\tfrac{1}{2}K_y \quad (z < 0)$$

Letting $\mathbf{a}_N$ be a unit vector normal (outward) to the current sheet, the result may be written in a form correct for all z as

$$\mathbf{H} = \tfrac{1}{2}\mathbf{K} \times \mathbf{a}_N$$

Note:

$$\mathbf{E} = \frac{\rho_S}{2\varepsilon_0}\mathbf{a}_N$$


Thank
you